

Methodological choices can outweigh algorithmic choices in transcriptomic immunotherapy prediction

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Abstract

Transcriptomic predictors of immune checkpoint inhibitor (ICI) response are benchmarked on small, incompatible cohorts with inconsistent endpoints, obscuring whether conclusions reflect algorithms or design. SPATBench isolates three such design choices across six methods and five immunotherapy-treated cohorts. Switching from a molecular surrogate (cytolytic activity) to a clinical endpoint (RECIST v1.1) in the same patients collapses every method whose features overlap the endpoint-defining genes (ElasticNet-Var 0.985 to 0.465) — a point-estimate effect larger than the largest within-cohort method gap — while two clinical definitions on identical patients move predictions by at most 0.083. The collapse replicates in executed cohorts spanning five cancer types. The Circularity Detection Score flags the circular endpoint far above its null, but its labels prove non-specific. No trainable method survives FDR correction at $n \leq 43$; the first detectable signal appears at $n = 73$. These results establish which benchmarking decisions, rather than which algorithms, determine conclusions.

Immune checkpoint inhibitors (ICIs) have transformed oncology [1,2], yet only 20–30% of unselected patients respond, motivating intensive efforts to develop transcriptomic predictors of response [3–6]. Multiple gene-expression-based methods have been proposed — the T-effector gene expression profile (GEP) [3], TIDE [4], IMPRES [5], and a growing set of machine learning (ML) models [6–8] — each evaluated on different cohorts, preprocessing pipelines, response definitions and protocols. The result is a literature in which nearly every method appears superior within its own developmental context [9,10]. Three structural features make method rankings unreliable here: (i) public immunotherapy cohorts with pre-treatment RNA-seq are small (typically $n < 50$) [11–13]; (ii) response can be defined by clinical endpoints (RECIST v1.1), molecular surrogates (cytolytic activity [14]), or progression-based endpoints (PFS), each capturing different biology; and (iii) design choices such as feature selection are rarely treated as experimental variables.

Recent benchmarks hold methodology fixed and compare methods across cohorts [15–18], and a re-analysis of seventeen tumor-microenvironment scoring methods found none reliable across cancer types [19]. I take the orthogonal approach: holding methods fixed and systematically varying methodological design choices, asking which design decisions — rather than which algorithms — determine benchmark outcomes. The question is not hypothetical: the immunophenoscore, a widely applied response surrogate, was derived by machine learning on TCGA data partitioned by median cytolytic activity (GZMA/PRF1)

within each cancer type [20], and a substantial literature trains predictors against TIDE-derived responder calls [21,22] (field audit in Results; Supplementary Table S30).

SPATBench, the framework and benchmark resource presented here, delivers four contributions: an open-source benchmark enforcing patient-stratified cross-validation and standardized preprocessing across six prediction methods and five immunotherapy-treated cohorts (196 patients; 169 melanoma); empirical isolation and quantification of three design choices — response definition, feature selection, and endpoint circularity — with effect-size comparisons against method-level variation; a reporting convention for pre-benchmarking endpoint assessment whose binding rule is the declaration of endpoint-defining genes, any gene-defined surrogate being circular by construction, with the Circularity Detection Score (CDS) as a null-referenced quantitative companion; and a complete permutation test matrix showing that no trainable method is distinguishable from label shuffling on any clinical endpoint at $n \leq 43$ after FDR correction, with minimum-sample-size standards derived from power analysis (Supplementary Table S31). All evidence is retrospective on public cohorts; the response-definition result is demonstrated in five cancer types via executed external replications.

RESULTS

Design Choice 1: Response Definition Exerts a Larger Effect Than Method Choice

The Riaz 2017 cohort provides two response definitions on the same patients and expression matrix.

Table 1. Riaz 2017: RECIST (n = 42) versus cytolytic (n = 43) response definitions (v33, HGNC-mapped). One patient lacked RECIST annotation and is excluded only from the RECIST endpoint.

Method	RECIST AUROC (n = 42)	Cytolytic AUROC (n = 43)	Δ AUROC	Type
ElasticNet-MI	0.569 [0.370–0.755]	0.968 [0.909–1.000]	−0.399	ML
ElasticNet-Var	0.465 [0.208–0.704]	0.985 [0.954–1.000]	−0.520	ML
TIDE	0.466 [0.222–0.723]	0.470 [0.282–0.653]	−0.004	Signature
IMPRES	0.613 [0.381–0.835]	0.660 [0.493–0.811]	−0.047	Signature
GEP	0.503 [0.305–0.710]	0.968 [0.915–1.000]	−0.465	Signature
PD-L1 (CD274 expression proxy)	0.545 [0.330–0.760]	0.868 [0.750–0.960]	−0.323	Biomarker

Switching response definitions transforms all ML methods from apparently state-of-the-art (ElasticNet-Var 0.985 on cytolytic) to near-chance on RECIST (0.465; Figure 1); GEP collapses with them (0.968 → 0.503) because the cytolytic endpoint is defined by GZMA/PRF1, both members of the GEP signature. IMPRES — whose fifteen checkpoint-gene relations share no genes with the endpoint definition — is nearly insensitive (0.660 → 0.613, $\Delta = -0.047$; permutation $p = 0.032$ on cytolytic, 0.154 on RECIST, each against its own null). Within the Riaz RECIST endpoint the largest between-method Δ AUROC is 0.148 (IMPRES vs ElasticNet-Var); the largest between-endpoint effect (0.520) exceeds it 3.5× in point estimate (Supplementary Table S14); the paired bootstrap difference is positive in 90% (frozen) and 84% (tuning) of resamples but is not separable from zero at 95% confidence at $n = 42$ (Methods). The contrast uses the seed-42 benchmark split; across 10 stratified seeds the $k = 2$ RECIST mean is 0.389 ± 0.105 , and re-running with $k = 3$ and $k = 5$ keeps all RECIST AUROC below 0.60 ($k = 3$: 0.397 ± 0.064 ; $k = 5$: 0.398 ± 0.088 ; Supplementary Table S16; cytolytic-endpoint seed variability in Supplementary Table S12) — the

collapse is not an artifact of the $k = 2$ constraint. Mechanistically, the cytolytic score is operationally defined by GZMA and PRF1 expression [14]: learning methods predict it from co-expressed immune-infiltration genes, and RECIST — not gene-defined — removes that leverage.

The effect replicates under pre-declared protocols in five cancer types through executed external replications (cohort characteristics, treatment classes, endpoints and responder fractions in Supplementary Tables S19, S20, S27, S28 and S29). On Liu 2019 (melanoma, $n = 122$, independent of all benchmark cohorts), the cytolytic surrogate yields GEP 0.959 and ElasticNet-MI 0.956 versus RECIST 0.535 and 0.538 ($\Delta = +0.425/+0.418$). On IMvigor210 (bladder carcinoma, $n = 297$), the largest Δ is 0.354 (GEP 0.967 $\rightarrow$ 0.613) against a same-cohort method gap of 0.180, with CDS separating the endpoints (0.953 vs 0.434). On GSE274975 (NSCLC, $n = 61$), the surrogate inflates GEP to 0.881 ($p = 2 \times 10^{-5}$) while PD-L1 remains the only fixed scorer surviving permutation on RECIST (0.690, $p = 0.014$). On IMmotion150 (renal cell carcinoma, $n = 165$), the surrogate yields GEP 0.905 (CDS 0.945) versus RECIST 0.662. On gastric cancer PRJEB25780 ($n = 78$), RECIST yields the strongest replication signal (GEP 0.744, BH $q = 0.001$) and the surrogate inflates GEP to 0.922. In every cohort the clinical endpoint's CDS sits below the surrogate's, and IMPRES — with no gene overlap — shows no surrogate inflation in any cohort.

The corresponding non-circular contrast is measured directly. Two public cohorts carry two clinical response definitions on the same patients. On Liu 2019, durable clinical benefit (DCB; progression-free at ≥ 6 months) versus RECIST moves predictions by at most 0.083 (ElasticNet-Var; $\kappa = 0.669$); on IMmotion150, the same construction moves predictions by at most 0.082 (IMPRES, 0.545 $\rightarrow$ 0.462) — both below the within-cohort method gaps (0.195 and 0.159) and roughly sixfold below the circular collapse (Supplementary Table S26). The collapse is therefore attributable to gene-defined surrogacy rather than to disagreement between progression-anchored clinical definitions.

Nor is the construction merely a hypothetical worst case: a structured audit (Supplementary Table S30) identified seven published studies training or selecting immunotherapy-response predictors against gene-expression-derived labels in cohorts without ICI outcomes — the immunophenoscore itself was derived against a cytolytic median split [20], and TIDE-derived responder calls serve as training labels across tumor types [21,22]. The design rules below therefore apply to this literature segment with the same force as to clinical-endpoint benchmarking.

Design Choice 2: Feature Selection Determines Biological Interpretation

Within identical ElasticNet models on Hugo 2016 ($n = 28$, RECIST), variance- and MI-based feature selection achieve indistinguishable performance and highly correlated out-of-fold predictions (Spearman $\rho = 0.75$), while their top-500 gene sets overlap by only 27% under the pipeline's two-stage prescreen convention (2.4% under a full-data MI convention — the overlap's dependence on the convention is itself a design-choice effect).

Table 2. ElasticNet on Hugo 2016 ($n = 28$): MI versus variance feature selection.

Metric	MI-based	Variance-based	Interpretation
AUROC (5-fold nested CV)	0.631	0.528	Not distinguishable (cross-seed $\Delta -0.008 \pm 0.050$, Table S12)
Gene overlap (top-500)	—	27.0% (135/500) prescreen convention; 2.4% (12/500) full-data MI	Overlap is convention-dependent

Metric	MI-based	Variance-based	Interpretation
Spearman ρ (OOF scores)	—	0.75 ($p < 0.001$)	Highly correlated predictions
Observed Δ AUROC (MI – Var)	+0.103 (benchmark split)	-0.008 ± 0.050 (10 seeds, Table S12)	Includes zero
Permutation p (5,000 full-pipeline shuffles; tested statistic = perm-obs, Table S11)	0.2853	0.3635	Neither distinguishable from chance

Neither variant is distinguishable from chance by permutation ($p = 0.2853$ and 0.3635 at $n = 28$; Table 2). At $n = 28$, feature selection changes the selected gene sets — and the biological narrative — without measurably changing predictive performance.

Design Choice 3: Endpoint Circularity Can Be Detected Before Benchmarking

The response-definition effect is mechanistically an instance of endpoint circularity: the cytolytic surrogate is operationally defined by genes inside the feature space. Design Choice 1 measured its magnitude within patients; Design Choice 3 asks whether circularity can be recognized before any benchmark is run.

Table 3. CDS with cohort-matched random-label nulls (200 replicates; declared genes GZMA/PRF1; v1.2.1). 'ML AUROC Range' = trainable methods only, v33 values. Riaz RECIST is reported on the HGNC matrix (0.409).

Endpoint	CDS	null mean	null max	CDS vs null	ML AUROC Range	Inflated?
Riaz 2017 cytolytic	0.954	0.566	0.842	far above null max	0.968–0.985	Yes
Gide 2019 RECIST	0.852	0.555	0.800	above null max	0.605–0.630	No
Lauss 2017 RECIST (ACT)	0.781	0.578	0.866	above null mean	0.627–0.700	No
Riaz 2017 RECIST (v33)	0.409	0.560	0.836	within null	0.465–0.569	No
Hugo 2016 RECIST	0.449	0.586	0.834	within null	0.528–0.631	No

CDS places the only circular endpoint far above its random-label null (Riaz cytolytic 0.954 vs null mean 0.566, max 0.842; Figure 2) — exactly where a GZMA/PRF1-defined endpoint must sit — while the clinical endpoints sit at or within their nulls (Hugo 0.449; Riaz RECIST 0.409; Lauss 0.781). The Gide RECIST value (0.852, above its null maximum) reflects genuine biology that the continuous score cannot separate from circularity: GEP reaches 0.830 ($q < 0.001$) there, and the executed replications return above-null readings as well. On real bulk-tumor data the composite reduces to an affine transform of two standard association statistics ($CDS \equiv 0.4 + 0.3 \cdot (C1n + C2)$; Methods), so the validated claims are the null-referenced ordering and the procedural rule that any gene-defined surrogate — of any program type (60 pseudo-endpoints in Supplementary Table S18, all HIGH) — is circular by construction; the shipped categorical labels are advisory. A retrospective consistency check supports this reading: had CDS existed in 2018, it would have returned 0.449 for the Hugo RECIST endpoint used in the Auslander et al. external validation of IMPRES — below its null mean, correctly indicating a non-circular endpoint.

Complete Permutation Test Matrix and Minimum Detectable Effects

Figure 3 shows the complete 36-cell permutation matrix (all 36 method-cohort cells carry a recorded permutation p ; within-cohort Benjamini–Hochberg families of six cells). On Hugo 2016, IMPRES alone survives (0.795, $q = 0.017$ — carrying the Carter et al. training-set-bias caveat [28]). On Gide 2019 ($n = 73$), GEP (0.830, $q < 0.001$), TIDE (0.752), PD-L1 (0.791), IMPRES and both ElasticNets (frozen-track 0.662/0.605, $q = 0.028$ each) survive — the first sample size at which a trainable method's signal becomes detectable on a clinical endpoint, a reading that is seed-stable across 10 regenerated stratified splits (ElasticNet-MI 0.613 ± 0.041 , ElasticNet-Var 0.624 ± 0.035 ; Supplementary Table S16). On the circular Riaz cytolytic endpoint even methods sharing the endpoint-defining genes survive (0.968–0.985) — the circularity effect itself. No trainable method survives FDR on any clinical endpoint at $n \leq 43$. Family-definition sensitivity is disclosed in Supplementary Table S21, and across cohorts the between-method spread reaches 0.508 (Jung 2019: GEP 0.786 vs TIDE 0.278) — which is why the endpoint effect must be demonstrated within-cohort. Across the four clinical-endpoint cohorts the methods swap ranks — IMPRES leads Hugo (0.795) and Riaz RECIST (0.613), GEP leads Gide (0.830), PD-L1 leads Lauss (0.780), and neither ElasticNet leads any column — so cross-cohort rankings conflate cohort effects with method effects.

AUROC alone does not capture clinical utility. On Hugo 2016, RECIST strongly stratifies overall survival (log-rank $p = 0.0003$) while IMPRES — the best AUROC method on that cohort — does not ($p = 0.554$). The same dissociation appears within the Gide significant cells themselves ($n = 73$; 29 deaths among OS-matched patients, OS from the cBioPortal Gide study): TIDE stratifies OS strongly (log-rank $p = 1.2 \times 10^{-6}$; Cox HR per SD 0.384) and PD-L1 likewise ($p = 3.4 \times 10^{-5}$; HR 0.53), whereas GEP — the best-AUROC method on this cohort — does not ($p = 0.055$), and IMPRES does not ($p = 0.56$); ElasticNet Brier and decision-curve metrics are reported in Supplementary Table S9b (results/benchmark/v33/gide_utility_metrics.json).

Power analysis sets the sample-size floor. At $n \leq 43$, 80% power for a single-sample AUROC comparison requires a true AUROC of 0.74–0.79, whereas at $n = 73$ it falls to 0.66 — the regime in which Gide becomes detectable. Pairwise $\Delta\text{AUROC} = 0.10$ comparisons require ≈ 130 patients under the most favorable correlation assumption (Supplementary Table S25). The resulting minimum-standards checklist (Supplementary Table S31) requires ≥ 1 clinical endpoint alongside any molecular surrogate, declaration of endpoint-defining genes, pre-specified multi-strategy feature selection with overlap reporting, non-learning reference baselines, full permutation testing with FDR correction, and ≥ 50 patients per cohort (≈ 130 for pairwise comparisons).

DISCUSSION

The three modules compose into one statement: a gene-defined surrogate misleads in proportion to (i) the immune-infiltration-driven co-expression supply linking endpoint genes to the feature space (high in melanoma, low in COAD-like contexts), (ii) the overlap between declared endpoint genes and the feature space (complete for the cytolytic surrogate, absent for IMPRES), and (iii) sample size, which sets the detectability floor. The TCGA anchor (2,491 patients; 12 immune signatures $\times$ 4 cancer types) is prognostic rather than predictive, and its composition confound is disclosed: the protective univariate ordering outside melanoma (SKCM 11/12 $\rightarrow$ BRCA 3/12 $\rightarrow$ LUAD 1/12 $\rightarrow$ COAD 0/12 significant signatures) is largely absorbed by age- and stage-composition under multivariate adjustment — now run for BRCA, LUAD and COAD (BRCA retains 1/12, Myeloid, adverse direction, HR 1.154, $q = 0.0011$;

LUAD and COAD 0/12; Table S15) — so it conditions rather than demonstrates the account, and the SKCM–COAD contrast partially conflates cancer type with metastatic-versus-primary composition. Schoenfeld-residual checks (rank test) on every multivariate model showed no proportional-hazards violation for the signature covariate in SKCM (0/12) or LUAD and COAD (0/12) and one marginal nominal violation in BRCA (Treg, $p = 0.0337$); the stage covariate violated non-proportionality in all SKCM and BRCA models. A stage-stratified Cox model (signature $z + \text{age } z$, stage-specific baseline hazards, no proportional-hazards assumption on stage) then reproduced every reported conclusion exactly (BH-significant signatures: SKCM 8/12, BRCA 1/12, LUAD 0/12, COAD 0/12), and a 24-month landmark analysis showed time-stable signature directions in SKCM (12/12); the single BRCA time-varying signature (Treg, harmful early HR 1.234, null late) is a non-significant row (data/tcga/ph_schoenfeld_check.json, data/tcga/stage_stratified_ph.json; Supplementary Table S15). The account also predicts that two clinical definitions of the same construct should barely move predictions; the RECIST-versus-DCB contrasts ($\Delta \leq 0.083$ across a melanoma and a renal-cell-carcinoma cohort) provide direct support.

SPATBench asks a different question from existing benchmarks [15–18]: which design choices cause benchmark unreliability at available sample sizes, and does the answer convert into design rules? The answer: response definition exerts a larger within-cohort effect ($\Delta = 0.520$) than any between-method difference on the same endpoint (0.148); across cohorts the between-method spread reaches 0.508 — so at $n = 43$, *how you define response can matter as much as or more than which algorithm you choose* (point estimate; paired bootstrap in Methods).

The genuinely new rules (Supplementary Table S31) are requiring a clinical endpoint alongside any gene-defined surrogate, declaring endpoint-defining genes, and the null-referenced companion reading; the remaining rows reinforce standards codified for prediction models and biomarker studies (REMARK, TRIPOD, PROBAST) [29–31]. One operational rule is not captured there: for endpoints with CDS > 0.80 — a conservative threshold, since genuine endpoints reach null maxima of 0.80–0.87 — assume ML methods are learning the endpoint-defining genes until independent clinical-endpoint validation proves otherwise. Feature selection deserves its own rule: accompany any biological interpretation with overlap reporting, the single most informative check for strategy dependence. The audit's clinical-development contrast matters here: the circular construction belongs to the surrogate-training lineage [20–22], while clinical development evaluates on clinical endpoints — which is why the two segments must be reported separately, and why the clinical-endpoint rule binds both. Beyond immunotherapy, any transcriptomic prediction task using a gene-defined surrogate carries the same risk, and the CDS concept generalizes to any expression matrix paired with a binary label, its validated reading remaining the null-referenced continuous value.

Method performance is inherently context-dependent [15,17]: IMPRES ranges from 0.795 (Hugo) to 0.583 (Jung DCB); GEP from 0.446 (Hugo) to 0.830 (Gide). Benchmark studies should report each method's known developmental limitations alongside performance [28]. Benchmark unreliability does not imply clinical uselessness — a predictor with wide CIs at $n = 28$ may still capture genuine biology that a larger cohort would confirm — but at current sample sizes one cannot *distinguish* reliable from unreliable benchmarks by AUROC alone, and the design choices documented here are the reason. Four boundaries qualify the scope of these conclusions.

1. All 95% CIs for trainable methods on clinical endpoints at $n \leq 43$ overlap 0.5; the sole fixed-scorer exception (IMPRES on Hugo) carries an inherited feature-selection-bias caveat [28]. No ML method is statistically superior at $n \leq 43$.
2. The collapse was demonstrated on Riaz and replicated in five cancer types with the same author-constructed surrogate; the remaining boundary is cohort availability. GSE274975 keeps its endpoints outside repository metadata, and IMmotion150's harmonization pools two treatment arms.
3. CDS's categorical labels lack specificity on real data (all 60 pseudo-endpoints score HIGH; 20–36% of real random-label replicates exceed the 0.70 boundary versus $\approx 3\%$ on the synthetic calibration null): the validated reading is the continuous value against a cohort-matched null, and circularity protection is procedural. CDS v2.0.0 adds a hypothesis-free mode and a declared-gene specificity statistic, validated across eight real endpoints: every circular cytolytic surrogate is flagged ($p = 0.005$ in all four), three of four clinical endpoints pass ($p \geq 0.09$), and the one clinical endpoint that fired (STAD RECIST, $p = 0.01$) is classified as non-circular by its declared-gene specificity (percentile 74.5, specificity $p = 0.11$); the permutation-based enrichment ratio itself proved fragile under permutation (the skew-ratio denominator collapses under label shuffling) and the v2 reading relies on the two new statistics.
4. The benchmark cohorts are predominantly melanoma (169/196) and one is adoptive T-cell therapy rather than ICI; Gide pools two treatment arms. TMB and PD-L1 IHC use expression proxies. Leave-one-cohort-out validation is precluded by gene-identifier heterogeneity — a barrier that is itself a finding.

METHODS

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Zhixian Huang (huangzhixian324@gmail.com), the sole author of this study.

Materials availability

This study did not generate new materials.

Data availability

- Data: All raw cohort data are publicly available from GEO (GSE78220, GSE91061, GSE100797, GSE135222, GSE274975) and cBioPortal (mel_iatlas_gide_2019, blca_iatlas_imvigora210_2017, mel_iatlas_liu_2019, rcc_iatlas_immotion150_2018) and ENA (PRJEB25780). Preprocessed expression matrices, all per-cell permutation outputs (results/benchmark/v33/nc_cells/ and nc_cells_ext/) and figures are archived in the versioned Zenodo record (DOI: 10.5281/zenodo.22746254).
- Code: All original code is publicly available. SPATBench pipeline (v33): <https://github.com/huangzhixian324-dev/spa-bench> (MIT). CDS package (v1.2.1): <https://github.com/huangzhixian324-dev/cds> (MIT); the packaged source is also included in the Zenodo archive (cds_tool/, installable with `pip install ./cds_tool`). A versioned archive with a

DOI is deposited on Zenodo (DOI: 10.5281/zenodo.22746254). The included Dockerfile builds the image (`docker build -t spa-bench:v33 .`) that reproduces the synthetic validation test (`scripts/validate.py`, exit code 0 verified) and ships the complete pipeline code; reproducing the reported tables additionally requires the Zenodo-archived inputs and the single documented entry point `scripts/rerun_v33.py`.

- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

Code availability

- Code: the pre-declared replication protocols are timestamped in the repository history — the Liu 2019 protocol and the GSE274975 pre-declared pipeline were committed at repository inception (2026-09-16, commit a745d2e), the IMmotion150 and STAD protocols on 2026-09-19, and the GSE274975 executed variant on 2026-09-20 (after endpoint recovery; Table S19); all replication outputs are archived in the same history and in the Zenodo record published 2026-09-22 (DOI: 10.5281/zenodo.22746254), which provides an independent third-party timestamp for the pre-declared protocols.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

This study is purely computational and generated no new experimental data: no experimental models, human participants, or animals were used. All cohorts are publicly available, de-identified bulk RNA-seq datasets from published studies (GEO accessions GSE78220, GSE91061, GSE100797, GSE135222 and GSE274975; cBioPortal studies `mel_iatlas_gide_2019`, `mel_iatlas_liu_2019`, `blca_iatlas_invigor210_2017` and `rcc_iatlas_immotion150_2018`; TCGA via the GDC), analyzed under the terms of their original publications. Sex- and gender-based analyses were not performed.

METHOD DETAILS

Cohort Selection and Preprocessing

Five publicly available cohorts with pre-treatment bulk RNA-seq data and annotated clinical outcomes were analyzed (Supplementary Table S3):

Cohort	Accession	Cancer	n	Treatment	Endpoint	Responders
Hugo 2016	GSE78220 [8]	Melanoma	28	Pembrolizumab	RECIST v1.1	13 (46%)
Riaz 2017	GSE91061 [7]	Melanoma	43 (42 with RECIST)	Nivolumab	RECIST + Cytolytic	9 RECIST (of 42) / 21 Cytolytic (of 43)
Lauss 2017 (ACT)	GSE100797 [34]	Melanoma	25	Adoptive TIL	RECIST v1.1	10 (40%)
Jung 2019	GSE135222 [35]	NSCLC	27	Anti-PD-1/PD-L1	DCB vs NDB (RECIST v1.1, ≥6-mo follow-up)	6 DCB (22%)
Gide 2019	<code>mel_iatlas_gide_2019</code> [36]	Melanoma	73	Anti-PD-1 ± anti-CTLA-4	RECIST v1.1	40 (55%)

Total: 196 patients (28 + 43 + 25 + 27 + 73). Cohorts were selected based on availability of pre-treatment bulk RNA-seq data with public clinical annotations. One cohort is adoptive T-cell therapy rather than checkpoint blockade: GSE100797 is the cohort of Lauss et al. (2017) [34] — referred to as "Nathanson" in parts of the benchmarking literature. The archived dataset key Nathanson_2017 is a legacy machine-readable alias retained for row-level reproducibility of the published tables, not an attribution. It is included deliberately as a stress test of method context-dependence beyond ICI; no headline claim depends on it, and I refer to the five cohorts as immunotherapy-treated throughout. The Riaz 2017 cohort provides two response definitions on the same patients — RECIST v1.1 from GEO SOFT metadata (of 49 pre-treatment biopsies with RECIST annotations, 9 had PR/CR and 33 had PD/SD; 7 had unknown response and were excluded, leaving $n = 42$ for the RECIST endpoint; the cytolytic endpoint uses all $n = 43$) and cytolytic activity (median-split of geometric mean of GZMA and PRF1; 21 above median [14]) — enabling a within-cohort, within-patient comparison of response definition effects. Cohorts were not pooled for meta-analysis due to endpoint heterogeneity (RECIST, DCB/PFS, and molecular surrogates capture different biology).

Label semantics: Throughout this work label 1 = clinical benefit (RECIST responder / DCB). For Jung 2019 (GSE135222), the original release coded 1 = NDB — the inverse of every other cohort; all Jung analyses use corrected labels (1 = DCB, 6/27).

All five benchmark cohorts' expression data were standardized to $\log_2(\text{FPKM} + 1)$; the executed external replications use cohort-appropriate normalization ($\log_2(\text{CPM} + 1)$ for the GSE274975 raw counts; $\log_2(\text{TPM} + 1)$ for the IMmotion150 iAtlas profile; Supplementary Tables S27 and S28). Patient-stratified k-fold cross-validation used $k = 5$ for all five cohorts except the Riaz RECIST endpoint, where the small responder count (9 of 42) constrained the design to $k = 2$. To assess k-fold robustness for the Riaz RECIST endpoint, I repeated the analysis with $k = 3$ and $k = 5$ (10 stratified random seeds each) and confirmed that the cytolytic→RECIST collapse magnitude is not an artifact of the k-fold choice (Supplementary Table S16; regenerated values in results/benchmark/v33/riaz_recist_kfold_v33.json). Gene symbols were mapped to HGNC where possible (Riaz 2017: Entrez-to-HGNC, 21,769/22,187 genes; Jung 2019: Ensembl-to-HGNC).

Methods Evaluated

Gene signatures (no training required): GEP (the 8-gene Ayers IFN- γ /T-effector signature: CD8A, GZMA, GZMB, IFNG, CXCL9, CXCL10, PRF1, TBX21 [3]; the circularity channel analyzed in this study is specific to gene sets containing GZMA/PRF1 — the 18-gene expanded GEP does not contain them and would not be expected to show the same collapse), TIDE (tidepy v1.3.9; the implementation was checked at source — $\text{classification} = \text{correlation}[\text{'TIDE'}] < \text{vthres}$, default $\text{vthres} = 0$ — consistent with its near-chance scores on both Riaz endpoints, Table 1), IMPRES (the 15 canonical pairwise checkpoint-gene relations of Auslander et al. [5]; the relations were decoded from the authors' published feature set (github.com/noamaus/IMPRES-codes, FEATS.mat) and cross-checked against the 9 pairs named in the paper text (9/9 match); the remaining 6 relations are not independently verified beyond the authors' released file. For a sample, the IMPRES score is the fraction of the 15 logical relations (e.g., $\text{CD40} > \text{PD-1}$) whose expression ordering holds, linearly rescaled when features are missing, exactly as described in the original Methods. No retraining is performed, mirroring the original design in which features were selected once on neuroblastoma spontaneous-regression data.)

Gene expression biomarker: PD-L1 (CD274 expression; proxy for IHC — real IHC scores unavailable in public metadata).

Machine learning (training required): ElasticNet logistic regression (sklearn saga solver, penalty = 'elasticnet') with variance-based feature selection (top-500 genes by expression variance in the training fold), ElasticNet with mutual information-based feature selection (top-500 genes by MI with the response label in the training fold), and XGBoost gradient boosting. ElasticNet hyperparameters (C, l1_ratio) were tuned via 3-fold inner cross-validation within each training fold; the selected hyperparameters were frozen during permutation testing, as is standard.

Supplementary baselines: TMB (constant 0.5; real mutation data unavailable) [37]. Immune deconvolution signatures (14 cell types, 43 unique genes; Supplementary Table S10) [38].

The six benchmarked prediction methods span four categories. All feature selection used only training-fold data.

Evaluation Protocol

Patient-stratified k-fold cross-validation with identical splits across all methods within each cohort. Ten metrics were computed. Here I report AUROC as the primary metric with 95% bootstrap confidence intervals (1,000 iterations) [39]. Expression-scale, preprocessing and cross-validation-strategy sensitivity analyses on Hugo 2016 (ElasticNet-MI) are provided in Supplementary Figure S5 and Tables S7 and S8. AUROC, AUPRC and bootstrap CIs for every method-cohort cell are provided in Supplementary Table S9, with clinical-utility metrics (Brier score, DCA, NRI, IDI) for a representative trainable cell; per-fold values are fully determined by the archived split files (Supplementary Table S1). For cohorts with $n \leq 28$, I additionally report cross-fold standard deviation (SD) as a stability metric.

Permutation Testing

Full permutation tests were performed for all trainable method-cohort pairs. Exact p-values = (number of permuted AUROCs $\geq$ observed + 1) / (n_permutations + 1). Benjamini-Hochberg FDR correction was applied within each cohort-endpoint family (six methods per family; sensitivity to alternative family definitions in Supplementary Table S21) [40].

Inclusion logic: Two protocols are used, matched to each method class. (a) Fixed scorers — IMPRES, GEP, TIDE, and PD-L1 (CD274 expression) — produce scores that are independent of the labels, so the full-pipeline label-shuffle null is mathematically identical to shuffling labels against the fixed out-of-fold scores; each cell therefore uses 50,000 prediction shuffles (seconds of compute, exact to 1/50,001). This equivalence was verified by running both protocols on Hugo IMPRES. (b) Trainable methods — ElasticNet-MI and ElasticNet-Var — are re-fit end-to-end under every label shuffle (per-fold feature selection with frozen hyperparameters), with a per-cell number of shuffles recorded in Supplementary Table S11. **XGBoost disclosure.** XGBoost is excluded from benchmark reporting (degenerate on all clinical endpoints; its only non-degenerate AUROC sits on the circular cytolytic endpoint — development-phase evaluation only). A completed permutation matrix for XGBoost across all six cohort-endpoints (1,000 full-pipeline shuffles per cohort-endpoint, default hyperparameters, the same patient-stratified folds as every other method) confirms the pattern with a non-ElasticNet learner: null on every clinical endpoint (Hugo p = 0.29, Nathanson p = 0.17, Jung p = 0.25, Riaz RECIST p = 0.81) and

significant only on the circular cytolytic endpoint (obs AUROC 0.903, $p = 0.0010$; Supplementary Table S16(e)). Re-examination (v33 protocol): under the current protocol XGBoost is not degenerate — re-runs produce non-degenerate predictions (Hugo 0.554, Lauss 0.493, Jung 0.611, Riaz-RECIST 0.434, Gide 0.718, and 0.991 on the circular cytolytic endpoint; all six cohorts re-run; results/benchmark/v33/xgb_exclusion_evidence_v33.json); it shows no signal on any clinical endpoint at $n \leq 43$, so the negative claim is unaffected. A 1,000-shuffle full-pipeline permutation test of the Gide re-run (frozen hyperparameters, the Table S11 protocol) gives $p = 0.004$ and $q = 0.007$ in the within-cohort family recomputed over seven cells, with the six original Gide q -values unchanged or more conservative. This re-run is reported as exploratory — not matrix-level — evidence: XGBoost does not enter the 36-cell matrix because the development-phase prediction tables were not retained under the current protocol, its permutation resolution (1,000 shuffles) falls below the Supplementary Table S31 floor for trainable cells, and its family structure (seven cells) differs from the pre-registered six-cell families; the detectability-onset reading at $n = 73$ is carried by the two pre-registered ElasticNet variants, and the XGBoost re-run is consistent with — but not constitutive of — that reading. The re-run and permutation evidence files are retained for audit (permutation record: results/benchmark/v33/nc_cells/Gide_2019_cBio__XGBoost.json; xgboost 3.4.1) even though the prediction tables are not.

Permutation resolution: p -values are exact given the recorded per-cell shuffle count: 50,000 for fixed scorers (minimum detectable $p = 2 \times 10^{-5}$) and 2,000–5,000 full-pipeline shuffles for the trainable methods, where the compute cost of re-fitting under every shuffle (≈ 60 – 90 s per shuffle) is the binding constraint. The initial budget ranged from 60 to 1,000 shuffles, under which the smallest achievable exact p at 60 shuffles ($1/61 \approx 0.016$) would have imposed a resolution ceiling inside a six-test cohort family. All eight clinical trainable cells were therefore extended to a uniform minimum of 2,000 shuffles (Hugo and Lauss MI/Var to 5,000; Jung and Riaz-RECIST MI/Var to 2,000) in a pre-registered resolution-hardening run (scripts/extend_nc_cells_5000.py, merged by scripts/merge_nc_cells_ext.py); no cell changed its significance verdict. The one borderline trainable cell, ElasticNet-MI on Gide 2019 ($n = 73$), had earlier been extended to 5,000 full-pipeline shuffles (checkpointed resumable protocol, frozen hyperparameters): the observed AUROC under this protocol is 0.662 (0.033 above the benchmark 0.629 — frozen-versus-tuned hyperparameters, the same deviation class disclosed in the Supplementary Table S11 footnote), giving $p = 0.0232$ and within-cohort BH $q = 0.028$ — **FDR-significant**. ElasticNet-MI therefore joins ElasticNet-Var as a detectable trainable signature on Gide ($n = 73$), strengthening the reading that Gide is the smallest sample size at which trainable transcriptomic signatures become detectable on a clinical endpoint — a reading that is seed-stable across 10 regenerated stratified splits (ElasticNet-MI 0.613 ± 0.041 , ElasticNet-Var 0.624 ± 0.035 ; Supplementary Table S16) and size-limited rather than cohort-limited: within-cohort subsampling of Gide keeps the ElasticNet-MI signal at a stable 0.62–0.66 mean AUROC from $n = 25$ to $n = 73$ while the 80%-power detection floor rises from 0.658 to 0.745, and detection remains marginal even at the full cohort size (2/10 draws at or above the floor; Supplementary Table S16); my headline claim is unchanged, since it is scoped to $n \leq 43$, where — at the extended resolution — no trainable method survives (the smallest within-cohort trainable q at $n \leq 43$ is 0.0716, Lauss ElasticNet-MI/Var; the smallest q on any cell is 0.058, Lauss PD-L1), independently corroborated by bootstrap 95% CIs that all overlap 0.5 (Supplementary Table S9). The frozen protocol itself was tested head-to-head against per-shuffle re-tuning on the Hugo ElasticNet-Var cell (300 label shuffles, both tracks from the pipeline-native nested CV): both protocols place the observed AUROC far inside their nulls ($p = 0.312$ frozen vs 0.259 re-tuned; null means 0.453 vs 0.425), so the negative claim is not an artifact of the frozen protocol (Supplementary Table S16).

Multiple Comparison Framework

Three levels of inference: (i) Exploratory: AUROC point estimates with 95% bootstrap CIs reported without multiplicity correction (descriptive quantities). (ii) Permutation testing: all evaluated method-cohort pairs with within-cohort FDR correction. (iii) Pairwise comparisons of AUROCs via paired bootstrap tests of Δ AUROC. The analytic DeLong covariance method ^[41] was considered but the bootstrap version was used throughout — I refer to these as paired bootstrap tests, not DeLong tests). The Bonferroni threshold ($p < 0.05/36 \approx 0.0014$) is unattainable for the trainable methods at their recorded shuffle resolutions — I treat this inaccessibility as a finding. Because family definition is an analysis degree-of-freedom, Supplementary Table S21 reports the full 36-cell matrix under three family definitions (within-cohort 6×6; global 36; method-class-stratified 24+12); the primary conclusions are family-robust, and the four family-dependent cells are disclosed there.

Effect-Size Comparison Framework

To compare the relative influence of methodological choices against method choice, I computed the range of AUROC values attributable to each source on the Riaz 2017 cohort: (a) between-endpoint Δ AUROC for each method; (b) between-method Δ AUROC within each endpoint; (c) between-feature-selection Δ AUROC within ElasticNet. I report these as effect-size comparisons rather than variance-component percentages, which are unreliable with limited independent data points. Between-method spreads differ across cohorts, so the endpoint effect must be compared with method effects at the same level of control. Within the same patients and the same RECIST endpoint (Riaz), the largest between-method Δ AUROC is 0.148 (IMPRES 0.613 vs ElasticNet-Var 0.465), so the largest between-endpoint effect (Var: 0.985 $\rightarrow$ 0.465, $\Delta = 0.520$) exceeds it by 3.5×; the same comparison yields $\approx 2.2\times$ on Liu 2019 (0.425/0.195) and $\approx 2.0\times$ on IMvigor210 (0.354/0.180) — the direction replicates, with a smaller ratio. The effect is further supported by non-overlapping bootstrap confidence intervals. These marginal per-endpoint intervals address the existence of the collapse (the Δ interval excludes zero) and ignore the within-patient correlation between the two AUROCs; the authoritative interval for the 3.5× comparison of the endpoint effect against the method gap is the paired difference reported above, which includes zero. The cytolytic-endpoint AUROC CIs for ElasticNet-Var ([0.954, 1.000]) and ElasticNet-MI ([0.909, 1.000]) do not overlap the RECIST-endpoint CIs ([0.208, 0.704] and [0.370, 0.755] respectively), and the same non-overlap pattern holds for the external replications (Liu 2019: cytolytic GEP [0.928, 0.986] vs RECIST [0.427, 0.642]; IMvigor210: cytolytic [0.948, 0.981] vs RECIST [0.536, 0.686]; IMmotion150: cytolytic [0.862, 0.942] vs RECIST [0.569, 0.750]); in GSE274975 ($n = 61$) the GEP intervals touch (cytolytic [0.777, 0.953] vs RECIST [0.432, 0.787]) while the point estimates separate cleanly. Across different cohorts the between-method spread can be as large as 0.508 (Jung: GEP 0.786 vs TIDE 0.278), so a cross-cohort comparison cannot support the claim; the controlled within-cohort comparison on Riaz is the appropriate design, because it fixes patients, treatment, and expression data and varies only the endpoint definition.

Uncertainty on the effect-size comparison. The 3.5× is a descriptive ratio of benchmark point estimates, not a bootstrap quantity. The inferential anchor is the Δ interval on the full 42-patient RECIST-evaluable set, which excludes zero under both hyperparameter constructions (frozen: $\Delta = 0.343$, 10,000-resample bootstrap 95% CI 0.090–0.598; tuning: $\Delta = 0.386$, CI 0.169–0.589; seed 42;). The paired difference between the endpoint effect and the method gap is positive in 90% (frozen) and 84% (tuning) of 1,000

resamples but is not separable from zero at 95% confidence (frozen: point 0.314, CI −0.121 to 0.420; tuning: point 0.240, CI −0.176 to 0.318;), so the ordering claim is calibrated to a point estimate throughout. Across different cohorts the between-method spread can reach 0.508, which is why the within-cohort comparison is the appropriate design.

Power Analysis

Minimum detectable Δ AUROC at 80% power ($\alpha = 0.05$, two-sided) as a function of sample size and responder proportion, using the Hanley–McNeil asymptotic variance of the AUC (Mann–Whitney U) for a paired two-method comparison on the same patients (baseline AUC 0.5) with AUC correlation $r = 0.75$ — the upper end of the correlation observed between method pairs on out-of-fold scores (Supplementary Table S12). The computation is archived as `scripts/power_table_v33.py` with a machine-readable sensitivity grid over $r \in \{0, 0.5, 0.75\}$ (`power_table_v33.json`); Supplementary Table S25 reports the $r = 0.75$ (most favorable) column, rounded to 0.01, and regenerates exactly from the archived script. Weaker correlation assumptions inflate the required n roughly two- to four-fold. Full power curves in Supplementary Figure S9.

Circularity Detection Score (CDS)

CDS quantifies endpoint-feature dependency as a weighted composite: **C1 (30%)**: mean MI between endpoint-defining genes and the response label, normalized as $\min(C1_MI / 0.05, 1)$. **C2 (30%)**: mean $|\text{Spearman } \rho|$ between endpoint-defining genes and the response label. **C3 (40%)**: $\min(1, R/10)$, where R is the ratio of the 90th-percentile MI of the top-500-variance features to the median MI of 500 random features. The composite is $CDS = 0.3 \cdot C1n + 0.3 \cdot C2 + 0.4 \cdot C3$. On real bulk-tumor cohorts $C3$ saturates at 1.000 (Limitations), so the released composite reduces to $CDS = 0.4 + 0.3 \cdot (C1n + C2)$ — an affine transform of two standard association statistics (mutual information and $|\text{Spearman } \rho|$); the validated reading is therefore the continuous value against a cohort-matched random-label null, not the weighting itself. These weights and normalization constants are those of the released tool (v2.0.0; source and packaging at <https://github.com/huangzhixian324-dev/cds> and in the Zenodo archive); every CDS value in this manuscript is exactly reproducible from them. Version 2.0.0 adds three statistics without changing the `v1.x` composite: a hypothesis-free mode that permutation-tests the genome-wide feature-label MI distribution without declared genes (an undeclared-surrogate detector), a declared-gene specificity statistic (the mean percentile rank of the declared genes within the genome-wide MI distribution, with a label-permutation null), and a label-permutation null for the $C3$ enrichment ratio. In validation across eight real endpoints, the hypothesis-free mode flagged every circular cytolytic surrogate ($p = 0.005$ in all four replication cohorts) and three of four clinical endpoints passed ($p \geq 0.09$); the one clinical endpoint that fired (STAD RECIST, $p = 0.01$) showed declared-gene specificity far below the circular ceiling (percentile 74.5 versus 100, specificity $p = 0.11$), demonstrating that association detection requires the specificity classifier. One calibration caveat applies wherever a categorical label is quoted: the HIGH boundary (0.70) was calibrated on synthetic nulls, where 3% of replicates exceed it, whereas 20-36% of real-cohort random-label replicates do (Limitations); the validated reading is the continuous value against its cohort-matched null (CDS scores are identical between v1.1.0 and v1.2.1 — v1.2.1 changes only the output semantics: it adds a null-referenced reading mode, a caveats block shipping the calibration limits below, and condition-specific guidance). The CDS concept builds on earlier work by Zhao et al. (2011, BMC Genomics), who introduced a "consistency degree" metric quantifying correlation between disease

endpoints and gene expression profiles [42]. I note two properties of this formulation: (a) C3 saturates at 1.0 for all evaluated endpoints and random-label replicates — highly variable genes in bulk tumor RNA-seq are predominantly immune-related, so the top-to-random MI ratio consistently hits the clipping ceiling regardless of endpoint circularity, and (b) I evaluated a re-weighting (C1 40% / C2 40% / C3 20%) as a candidate refinement, but the released tool and all reported values use the 30/30/40 formulation. Permuting C3's role (v2.0) is planned to improve MODERATE-range discrimination. **Synthetic data calibration:** To validate CDS independently of my three real endpoints, I generated synthetic expression data ($n = 200, 5,000$ genes) with controlled circularity levels (0–100% endpoint-gene signal, with 50 independent biological-signal genes as non-circular signal). CDS ranks the five scenarios monotonically ($\text{CDS } 0.697 < 0.820 < 0.830 < 0.832 < 0.845$; Supplementary Table S4d). Scenarios with $\text{CDS} \geq 0.80$ show inflated AUROC (0.691–0.845), while the pure-random scenario scores 0.697 — above the level of an unstructured label, reflecting C3 saturation.

Retrospective external consistency check: The Auslander et al. (2018) study provides an independent validation context for CDS. In that study, IMPRES was trained on Auslander's own data and validated externally on Hugo 2016 (RECIST endpoint), achieving AUROC = 0.83 in the original paper (0.795 in my benchmark). If CDS had been computed before that validation was performed, it would have returned $\text{CDS} = 0.449$ (MODERATE under the shipped thresholds, and below its cohort-matched random-label null mean of 0.586), indicating that the Hugo RECIST endpoint is not circular — any observed AUROC reflects genuine biological signal in the IMPRES gene pairs, not endpoint-gene dependency. This counterfactual application of CDS to a published external validation scenario constitutes a retrospective consistency check: CDS's reading (non-circular endpoint $\rightarrow$ AUROC not inflated by endpoint-gene dependency) is consistent with the observed outcome (IMPRES achieved significant AUROC = 0.795 on Hugo RECIST, permutation $p = 0.0029$). The same logic applies to Lauss 2017 (ACT) ($\text{CDS} = 0.781$, MODERATE under the recalibrated boundary; no trainable AUROC exceeds 0.70, no inflation). For Gide 2019 no CDS value was stored at analysis time; the v33 recomputation (Supplementary Table S4b) returns $\text{CDS} = 0.852$ — the C1/C2 components read the strong association between the declared genes (GZMA/PRF1) and the clinical label in this large melanoma cohort as endpoint-gene dependency, i.e. the known inability of CDS v1.x to separate genuine immune-biology signal (GEP AUROC = 0.830, $q < 0.001$) from circularity. CDS correctness claims are therefore restricted to the endpoints with stored scores.

Null distribution, stress test, and reporting convention: I generated 30 independent replicates of random binary labels to estimate the CDS null distribution. Mean null CDS = 0.50 (SD = 0.09), 95th percentile = 0.65, 99th percentile = 0.71. Approximately 3% of random endpoints were flagged as HIGH under the shipped threshold ($\text{CDS} > 0.70$) due to C3 saturation at 1.0 — a known limitation of the current C3 formulation; no random endpoint achieved $\text{CDS} < 0.30$. I disclose that these 30 null replicates were generated **after** the four real benchmark endpoints had been inspected; threshold calibration is therefore retrospective, not prospective, and categorical risk labels must be treated as advisory. I deliberately ship a single threshold set (HIGH > 0.70 , MODERATE 0.30–0.70, LOW ≤ 0.30) — identical in the manuscript, in the released pip package, and in all archived result files — so that users replicating my analysis obtain exactly my labels. A more conservative HIGH threshold of 0.80 (false-positive rate 0/30 on the synthetic null) is reported for transparency and will be adopted in CDS v2.0 together with a permutation-based C3. I recommend reporting the continuous CDS value; categorical risk assignments are advisory.

To characterize what CDS measures, I generated 30 pseudo-endpoints per cohort from real gene programs [43] — ECM/stromal (collagens, fibroblast, endothelial, pericyte and basement-membrane markers), tumor-purity/proliferation (cell cycle, E2F/MYC targets, glycolysis, oxidative phosphorylation) and lineage/signalling programs (melanocytic differentiation, keratinocyte basal, hypoxia, interferon, antigen presentation) — each dichotomized at the median of the program-mean expression, the same construction as the cytolytic surrogate, with the program's own genes declared as endpoint genes. All 60 pseudo-endpoints (Gide 2019, $n = 73$; Hugo 2016, $n = 28$) scored above the HIGH boundary (CDS 0.792–0.946): CDS flags any operationally gene-defined surrogate regardless of program identity. Random-label nulls computed per endpoint on the real feature spaces (10 replicates each; Table 3) show that the shipped 0.70 threshold also captures 3/10 null replicates at $n = 28$. CDS is therefore validated as a continuous, null-referenced measure of endpoint–gene association; the categorical HIGH label carries no specificity among structured endpoints, and the protection it enables is procedural — declaring the endpoint-defining genes identifies the circular construction itself. Full design and per-endpoint results in Supplementary Table S18 and Note S6. CDS values are reported alongside the mean, maximum and 95th percentile of a 200-replicate random-label null computed on the same cohort and feature space (Table 3); the archived 10-replicate nulls give the same readings for every endpoint. A CDS far above the null maximum indicates that the declared genes track the label beyond chance — expected for operationally gene-defined endpoints (circularity), but also possible for genuine biology when the declared genes are true correlates of response (the Gide 2019 case, Table S4b). The two are distinguishable only by endpoint provenance, which is why the declaration of endpoint-defining genes is the primary control. **Epistemological note:** I recognize the inherent tension: CDS is itself subject to the same validation standards I recommend for benchmarking tools — prospective evaluation on independent multi-endpoint cohorts, transparent reporting of limitations, and third-party replication. Endpoint–feature overlap of the kind CDS detects is one of the leakage classes formalized in the machine-learning reproducibility literature [44]. To facilitate this, CDS is released as an open-source, independently installable package with a documented API and bundled synthetic calibration data. Its current status is experimental, and its provisional thresholds should be refined as more multi-endpoint cohorts become available.

IMPRES Implementation

IMPRES uses the canonical implementation decoded from the original authors' released feature set: 15 fixed logical relations (list in Supplementary Note S5), score = fraction fulfilled, no retraining. This matters because IMPRES on Hugo 2016 is the headline positive result of this study — it is the only FDR-significant non-circular-endpoint result at $n \leq 43$ (Supplementary Table S11); that result must rest on the published algorithm, not on an ad hoc variant. On the canonical implementation, IMPRES on Hugo 2016 yields AUROC = 0.795 (permutation $p = 0.0029$ at 50,000 prediction shuffles under the fixed-scorer protocol; $q = 0.017$ after BH). I note that Carter et al. (2019) identified potential training-set bias in the original IMPRES feature selection (34% non-random repeats) [28]. The extent to which IMPRES's Hugo performance reflects genuine biology versus inherited

feature selection bias is unresolved; I report the AUROC as observed and flag this interpretative caveat.

QUANTIFICATION AND STATISTICAL ANALYSIS

Permutation testing uses full-pipeline label shuffles with frozen hyperparameters; exact p-values are $(ge + 1)/(n + 1)$. Within-cohort multiple testing uses Benjamini-Hochberg FDR ($q < 0.05$). AUROC 95% CIs use the paired bootstrap (1,000 iterations, the convention used throughout and in Table S9); correlated-AUROC comparisons likewise use the paired bootstrap rather than DeLong. Power analysis and minimum-sample guidance are derived from observed effect sizes (Supplementary Table S9).

ADDITIONAL RESOURCES

None.

SUPPLEMENTARY DATA

Supplementary Information is provided as a separate PDF (Markdown source `supplementary_nmi_v1.md`, assembled from the machine-readable outputs in `results/benchmark/v33/`). Contents:

- **Table S1:** Per-fold cross-validation setup
- **Table S2:** Literature comparison — published vs. SPATBench AUROC
- **Table S3:** Cohort characteristics (detailed)
- **Table S4:** CDS — full component breakdown, synthetic calibration data, null distribution analysis, recalibrated thresholds
- **Table S5:** Method concordance — Cohen's κ , full 10-pair matrix (Hugo 2016)
- **Table S6:** IMPRES leave-one-pair-out analysis
- **Table S7:** Preprocessing sensitivity
- **Table S8:** Cross-validation sensitivity
- **Table S9:** Full five-cohort benchmark results (AUROC/CI/AUPRC, clinical-utility metrics, survival stratification)
- **Table S9b:** Gide 2019 clinical-utility metrics for the FDR-significant cells
- **Table S10:** Immune deconvolution marker genes (43 genes, 14 signatures)
- **Table S11:** Complete permutation test results (all 36 cells, including the frozen-pipeline observed AUROC)
- **Table S12:** Random seed stability analysis (10 seeds)
- **Table S13:** Learning curve analysis (Hugo 2016, $n = 8-28$)
- **Table S14:** Effect-size comparison — Riaz 2017
- **Table S15:** TCGA Cox regression (12 signatures $\times$ 4 cancer types, 2,491 patients; COAD re-acquired from the GDC current index)
- **Table S16:** k-fold and seed robustness — Riaz RECIST k-fold (a); Gide RECIST 10-seed rerun (b)
- **Table S17:** CDS retrospective assessment on five published benchmark studies + computed CDS for the iAtlas-accessible cohorts (E3)
- **Table S18:** CDS negative-control stress test — 60 real gene-program pseudo-endpoints (E2)

- **Table S19:** External replication — response-definition effect on Liu 2019 (independent cohort, E1)
- **Table S20:** Cross-cancer replication — IMvigor210 (bladder carcinoma, non-melanoma, E1)
- **Table S21:** Multiple-comparison family sensitivity — BH q under within-cohort / global / method-class-stratified family definitions (36 cells)
- **Table S22:** AUROC across four clinical-endpoint cohorts (migrated from the main text; 95% CIs where shown)
- **Table S23:** DCB endpoint, Jung 2019 (n = 27, NSCLC; corrected labels)
- **Table S24:** Overall survival stratification on Hugo 2016 (true events)
- **Table S25:** Minimum detectable Δ AUROC at 80% power (Hanley–McNeil paired model, $r = 0.75$)
- **Table S26:** Clinical–clinical response-definition contrast on Liu 2019 (RECIST vs PFS-derived DCB)
- **Table S27:** Executed NSCLC replication — GSE274975 (Poddubskaya 2024)
- **Table S28:** Fourth-cancer-type replication — IMmotion150 ccRCC (McDermott 2018)
- **Table S29:** Fifth-cancer-type replication — gastric cancer PRJEB25780 (TIGER portal)
- **Table S30:** Field audit — published studies training or selecting immunotherapy-response predictors against gene-expression-derived response labels
- **Table S31:** Minimum requirements for reliable immunotherapy prediction benchmarking
- **Figures S1–S12**
- **Notes S1–S6**

FIGURE LEGENDS

Figure 1. Response-definition collapse (Riaz 2017, HGNC-mapped, v33; 95% bootstrap CIs)

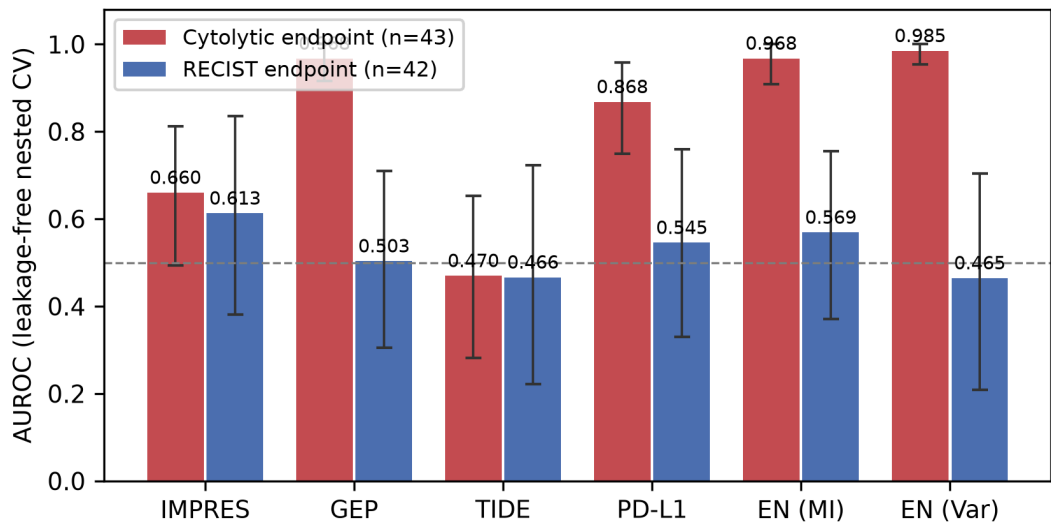


Figure 1. Response-definition collapse on Riaz 2017. AUROC per method on the same patients under the cytolytic-surrogate endpoint versus the clinical RECIST endpoint (HGNC-mapped matrix, v33; 95% bootstrap CIs). Every method whose feature set overlaps the endpoint-defining genes (GZMA/PRF1) collapses; IMPRES, with no overlap, is nearly insensitive. The identical file is provided as Supplementary Figure S1.

Alt text: Bar chart comparing AUROC of six prediction methods on the same Riaz 2017 patients under two endpoints: the gene-defined cytolytic surrogate (all methods with overlapping feature sets score high,

0.83-0.99) versus the clinical RECIST endpoint (all collapse to 0.40-0.57). IMPRES, whose feature set does not overlap the endpoint genes, is nearly unchanged between endpoints.

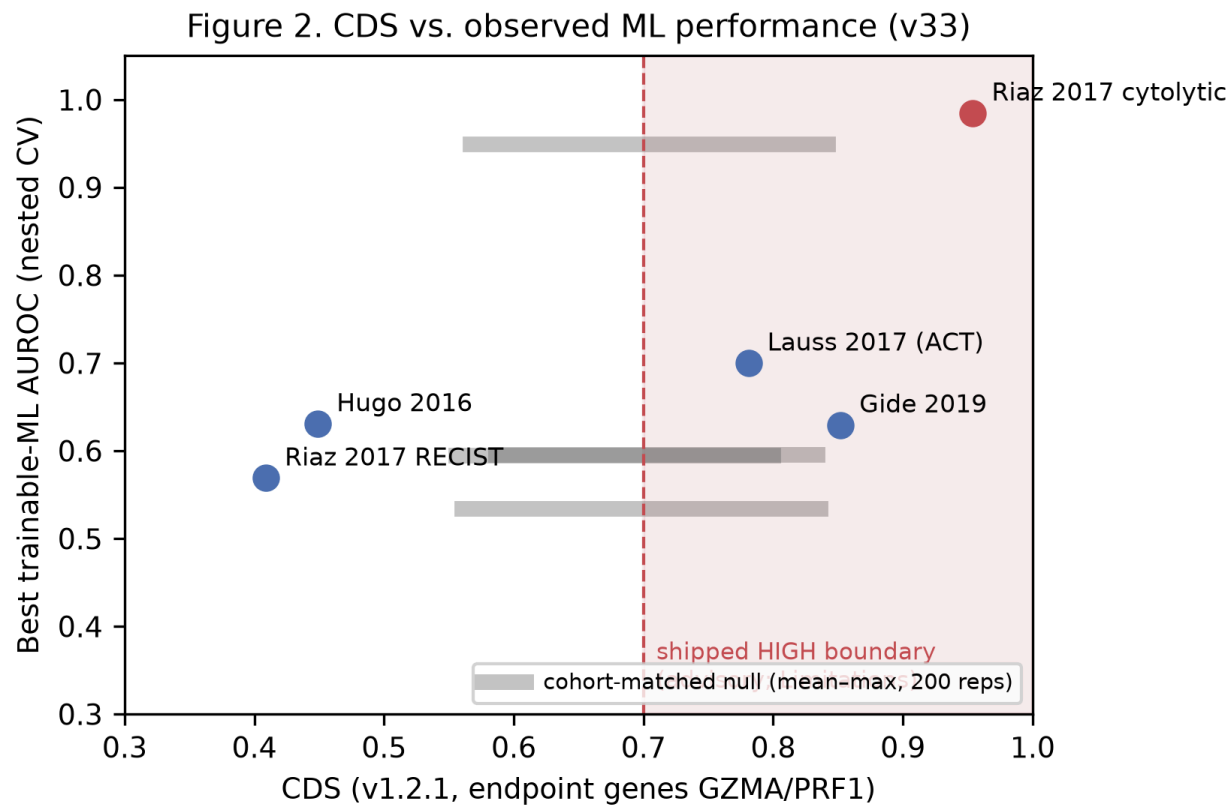


Figure 2. CDS versus observed trainable-ML performance. Circularity Detection Score (v1.2.1; scores identical to the v1.1.0 recomputation, Table S4b) against the best trainable-ML AUROC per endpoint, with cohort-matched random-label nulls; the circular cytolytic endpoint separates from clinical endpoints. The identical file is provided as Supplementary Figure S10.

Alt text: Scatter plot of Circularity Detection Score versus best trainable-ML AUROC across five endpoints, with cohort-matched random-label nulls. The circular cytolytic endpoint (CDS 0.954) separates from the clinical endpoints, which sit at or inside their nulls except Gide RECIST (CDS 0.852, above its null maximum, reflecting genuine biology the categorical label cannot separate).

Figure 3. Complete permutation matrix (E4): AUROC + p per cell; * BH $q < 0.05$ within cohort. Fixed scorers: 50,000 prediction shuffles; trainable: full-pipeline shuffles with frozen hyperparameters (per-cell n); trainable cells show obs/perm-obs (p tests perm-obs)

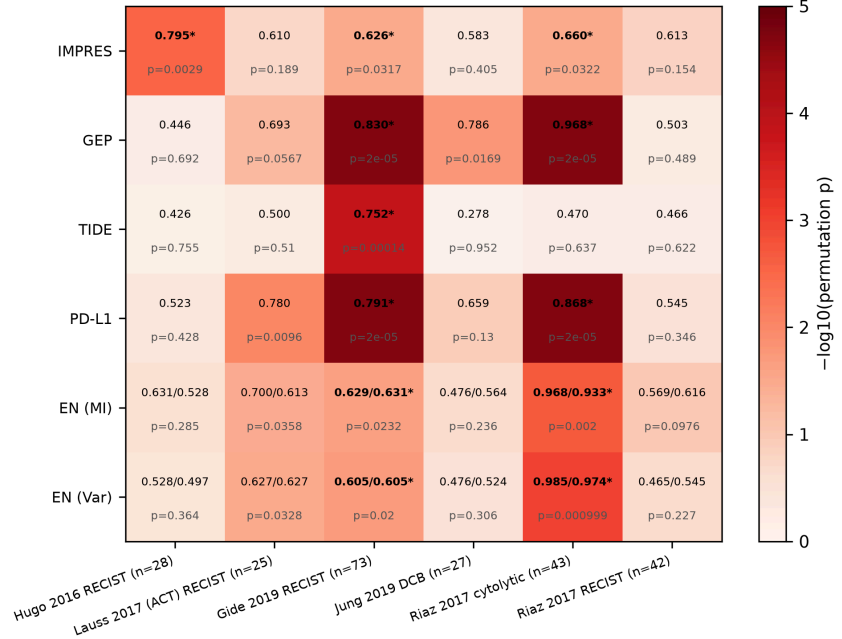


Figure 3. Complete permutation test matrix. Observed AUROC and permutation p per method–cohort cell; trainable cells show obs/perm-obs (the p value tests the perm-obs statistic under frozen hyperparameters, Table S11); asterisks mark within-cohort BH $q < 0.05$. No trainable method survives FDR on any clinical endpoint at $n \leq 43$. The identical file is provided as Supplementary Figure S12.

Alt text: Heatmap of the complete 36-cell permutation test matrix showing observed AUROC (trainable cells: obs/perm-obs) and permutation p per method-cohort cell; asterisks mark within-cohort BH q below 0.05. No trainable method survives FDR correction on any clinical endpoint with n at most 43.

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AUTHOR CONTRIBUTIONS

The sole author was responsible for all aspects of this work: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Resources, Data curation, Writing - original draft, Writing - review and editing, Visualization, Supervision, Project administration, and Funding acquisition.

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DECLARATION OF INTERESTS

The author declares no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE WRITING PROCESS

During the preparation of this work the author used ChatGPT (OpenAI) in order to improve the clarity and language of the manuscript. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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